MSG No. 80.32 I_c4_1 [Type IV, tetragonal]

Table 1: Wyckoff site: `8a`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[1,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{4}\right]$	`[2,15]`
3	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{4}\right]$	`[3,14]`
4	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[4,13]`
5	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[5,12]`
6	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[6,11]`
7	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{3}{4}\right]$	`[7,10]`
8	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[8,9]`

Table 2: Wyckoff site: `8b`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{4}\right]$	`[2,15]`
3	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{4}\right]$	`[3,14]`
4	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[4,13]`
5	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[5,12]`
6	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[6,11]`
7	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{3}{4}\right]$	`[7,10]`
8	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[8,9]`

Table 3: Wyckoff site: `8c`, site symmetry: `2..`

No.	position	mapping
1	$[0, 0, z]$	`[1,4]`
2	$\left[0, \frac{1}{2}, z+\frac{1}{4}\right]$	`[2,3]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[5,8]`
4	$\left[\frac{1}{2}, 0, z+\frac{3}{4}\right]$	`[6,7]`
5	$\left[0, 0, z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[0, \frac{1}{2}, z+\frac{3}{4}\right]$	`[10,11]`
7	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[13,16]`
8	$\left[\frac{1}{2}, 0, z+\frac{1}{4}\right]$	`[14,15]`

Table 4: Wyckoff site: `16d`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$\left[-y, x+\frac{1}{2}, z+\frac{1}{4}\right]$	`[2]`
3	$\left[y, \frac{1}{2}-x, z+\frac{1}{4}\right]$	`[3]`
4	$[-x, -y, z]$	`[4]`
5	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	`[5]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{2}-y,\, x,\, z+\frac{3}{4}\right]$	[6]
7	$\left[y+\frac{1}{2},\, -x,\, z+\frac{3}{4}\right]$	[7]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[8]
9	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[9]
10	$\left[-y,\, x+\frac{1}{2},\, z+\frac{3}{4}\right]$	[10]
11	$\left[y,\, \frac{1}{2}-x,\, z+\frac{3}{4}\right]$	[11]
12	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[13]
14	$\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{4}\right]$	[14]
15	$\left[y+\frac{1}{2},\, -x,\, z+\frac{1}{4}\right]$	[15]
16	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[16]